

Logic: Coverage Criteria

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Re-cap: Predicates and clauses

- A predicate is an expression that evaluates to a Boolean value.
- E.g.: $(x > y) \vee C \vee f(z)$ is a predicate where x, y and z are non-Boolean variables, C is a Boolean variable and $f(z)$ is a function that returns a Boolean constant.
- A clause is a predicate that does not contain any logical operators.
- E.g., in the above predicate, there are three clauses: $x > y$, C and $f(z)$.

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Equivalent logical expressions

- There could be several different ways of writing a predicate such that all of them are syntactically equivalent and hence, are semantically also equivalent.
- While working with coverage criteria, the criteria are defined based on the syntax sometimes.
- We will assume that a given predicate yields the same test requirements for a given coverage criterion no matter which form of the logical expression is used.

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Predicates and clauses: Notations

- Let P be a set of predicates and C be a set of clauses in the predicates in P.
- For each predicate $p \in P$, let C_p be the clauses in p. $C_p = \{c \mid c \in p\}$.
- $C = \cup_{p \in P} C_p$.

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Predicate coverage

- Predicate coverage (PC): For each $p \in P$, TR contains two requirements: p evaluates to true and p evaluates to false.
- E.g., for the predicate $(x > y) \vee C \vee f(z)$, the two tests that satisfy predicate coverage will be $x = 5, y = 3, C = \mathit{true}, f(z) = \mathit{false}$ (the predicate evaluates to true) and $x = 1, y = 4, C = \mathit{false}, f(z) = \mathit{false}$ (the predicate evaluates to false).
- Overlap with graph coverage criteria: For a set of predicates associated with branches, predicate coverage is the same as edge coverage.

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Clause coverage

- Individual clauses are not *exercised* in predicate coverage.
- For e.g., in the predicate $(x > y) \vee C \vee f(z)$, the clause $f(z)$ was false in both the tests.
- Clause coverage (CC): For each $c \in C$, TR contains two requirements: c evaluates to true and c evaluates to false.
- For e.g., clause coverage in our predicate $(x > y) \vee C \vee f(z)$, needs each clause to be made true once and false one. Tests for clause coverage could be
  $x = 5, y = 3, C = \mathit{true}, f(z) = \mathit{true}$ and
  $x = 5, y = 6, C = \mathit{false}, f(z) = \mathit{false}$.

Predicate coverage vs. clause coverage

- Clause coverage does not subsume predicate coverage.
- Consider $p = a \vee b$. The clauses are a and b. There are four test inputs that consider all combinations of true/false values for a and b.

	a	b	$a \vee b$
1	T	T	T
2	T	F	T
3	F	T	T
4	F	F	F

- The test set $\{2, 3\}$ above satisfies clause coverage but the predicate is true in both the cases. Hence, predicate coverage is not satisfied.

Combinatorial coverage

- Combinatorial coverage (CoC): For each $p \in P$, TR contains test requirements for the clauses in C_p to evaluate to each possible combination of truth values.
- Combinatorial coverage is commonly called as multiple condition coverage.
- For the predicate $p = a \vee b$, combinatorial coverage will test all the values in the truth table below.

	a	b	$a \vee b$
1	T	T	T
2	T	F	T
3	F	T	T
4	F	F	F

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Combinatorial coverage: Feasibility

- A predicate p with n clauses has 2^n possible assignments of truth values.
- Hence, combinatorial coverage is many times not feasible.
- In fact, combinatorial coverage may not be needed most of the times.
- We need criteria that capture the effect of each clause on its predicate and to do this within a reasonable number of tests.

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Clause determining a predicate

- Sometimes, the truth of certain clauses in a predicate makes the other clauses non-influential on the predicate.
  - For e.g., in the predicate $p = a \vee b$, if a is true then the predicate p is true irrespective of whether b is true or false.
  - If a is false, then b completely *determines* p in the sense that truth/falsity of b influences truth/falsity of p.
- We would like to test *each* clause under circumstances where the clause determines the predicate.

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Clause determining a predicate

- At a given point in time, we are interested in one clause in a predicate: we call this the major clause.
- All other clauses are minor clauses.
- Each clause, in turn, is treated as a major clause.
- Determination: Given a major clause c_i in predicate p, we say that c determines p if the minor clauses $c_j \in C_p$, $j \neq i$, have values so that changing the truth value of c_i changes the truth value of p.
- Note: This *does not* require that $c_i = p$. It also does not insist that the truth value of p is the same as the truth value of c_i.

Active clause coverage

- Active Clause Coverage (ACC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses $c_j, j \neq i$, so that c_i determines p.
  TR has two requirements for each c_i: c_i evaluates to true and c_i evaluates to false.
- For e.g., if $p = a \vee b$, we get four requirements in TR, two for clause a and two for clause b. b needs to be false for a to determine p and vice versa.

	a	b
$c_i = a$	**T**	F
	F	F
$c_i = b$	F	**T**
	F	**F**

- There are three distinct TRs above:
  $\{(a = T, b = F), (a = F, b = T), (a = F, b = F)\}$.

Active clause coverage

- Active clause coverage chooses different clauses to be major clause in turn.
- TR has requirements for each clause as a major clause.
- For a predicate with n clauses, $n + 1$ distinct test requirements suffice to achieve active clause coverage.

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Modified Condition Decision Coverage

- Modified Condition Decision Coverage (MCDC) criterion is a criterion that states each clause in a predicate be made true and false in turn and the predicate be tested for each clause this way.
- This is the same as our active clause coverage.
- MCDC criterion is mandatory by several certification standards for safety critical software.

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Active clause coverage: Moving on

- In active clause coverage, we don't consider the values that the minor clauses take.
- For a small predicate like $p = a \vee b$, this will not matter, but it does for predicates with more clauses.
- The minor clauses could all take the same value or could take different values.
- Three kinds of active clause criteria are defined based values that the minor clauses take: General active clause coverage, Correlated active clause coverage and Restricted active clause coverage.

General active clause coverage

- General Active Clause Coverage (GACC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses c_j, $j \neq i$ such that c_i determines p.
  TR has two requirements for each c_i: c_i evaluates to true and c_i evaluates to false. The values chosen for the minor clauses c_j *do not* need to be the same when c_i is true as when c_i is false.
- GACC *does not* subsume predicate coverage.
  - For e.g., consider $p : a \equiv b$. Clause a determines p for any value of b.
  - We choose the test inputs $\{a = b = T, a = b = F\}$. p evaluates to true for both the cases, so predicate coverage is *not* achieved.

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Correlated active clause coverage

- Correlated Active Clause Coverage (CACC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses c_j, $j \neq i$ such that c_i determines p.
  TR has two requirements for each c_i: c_i evaluates to true and c_i evaluates to false. The values chosen for the minor clauses c_j must cause p to be true for one value of the major clause c_i and false for the other.
- For the example $p : a \equiv b$, CACC tests are:
  - a is the major clause: Test set $\{TT, FT\}$ (TT to be read as $a = T, b = T$).
  - b is the major clause: Test set $\{TT, TF\}$.
  - One is common. Merging, we get the CACC test set as $\{TT, TF, FT\}$.
- In fact, CACC subsumes predicate coverage.

Restricted Active Clause Coverage

- Restricted Active Clause Coverage (RACC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses c_j, $j \neq i$ such that c_i determines p.
  TR has two requirements for each c_i: c_i evaluates to true and c_i evaluates to false. The values chosen for the minor clauses c_j must be the same when c_i is true as when c_i is false.

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ACC Criteria: Example

- We illustrate the various ACC criteria using another example.
- Consider the predicate $p = a \wedge (b \vee c)$.
- Truth table for p.

	a	b	c	$a \wedge (b \vee c)$
1	T	T	T	T
2	T	T	F	T
3	T	F	T	T
4	T	F	F	F
5	F	T	T	F
6	F	T	F	F
7	F	F	T	F
8	F	F	F	F

CACC on $p = a \wedge (b \vee c)$

- In $p = a \wedge (b \vee c)$, for a to determine p, $b \vee c$ must be true.
- $b \vee c$ can be made true in three ways.
- CACC is satisfied for a being the major clause by choosing one TR from rows 1,2,3 and one from rows 5,6,7.

	a	b	c	$a \wedge (b \vee c)$
1	T	T	T	T
2	T	T	F	T
3	T	F	T	T
5	F	T	T	F
6	F	T	F	F
7	F	F	T	F

- Totally, nine combinations exist.

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RACC on $p = a \wedge (b \vee c)$

- Clause a is the major clause: Only three of the nine sets of test requirements that satisfy will satisfy RACC.
- Row 1 is paired with row 5, row 2 is paired with row 6 and row 3 is paired with row 7.

	a	b	c	$a \wedge (b \vee c)$
1	T	T	T	T
5	F	T	T	F
2	T	T	F	T
6	F	T	F	F
3	T	F	T	T
7	F	F	T	F

Logic coverage criteria seen so far

- Predicate coverage.
- Clause coverage.
- Combinatorial coverage.
- Active clause coverage.
  - Generalized active clause coverage.
  - Correlated active clause coverage.
  - Restricted active clause coverage.

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Inactive clause coverage

- Complementary criterion to active clause criteria, ensures that the major clause *does not* affect the predicate.
- Inactive Clause Coverage (ICC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses $c_j, j \neq i$ so that c_i *does not* determine p.
  TR has four requirements for c_i:
  1. c_i evaluates to true with p true.
  2. c_i evaluates to false with p true.
  3. c_i evaluates to true with p false.
  4. c_i evaluates to false with p false.

Why ICC?

- You might wonder why ICC?
- Here is a simple example justifying ICC:
  - System: Control software for a shutdown system in a reactor. Two modes: Normal and Override.
  - Specification: The status of a particular valve (open or closed) is relevant to the reset operation in Normal mode but not in Override mode.
  - Reset should behave identically in Override mode when the valve is open and when it is closed.
- ICC criterion for reactor: Override mode does not affect the reset.

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Extensions of ICC

- Like done for ACC, ICC is detailed out into general and restricted ICC.
- Correlated ICC does not make sense as the major clause does not determine the predicate.

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General Inactive Clause Coverage

- General Inactive Clause Coverage (GICC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses $c_j, j \neq i$ so that c_i *does not* determine p.
  TR has four requirements for c_i:
  1. c_i evaluates to true with p true.
  2. c_i evaluates to false with p true.
  3. c_i evaluates to true with p false.
  4. c_i evaluates to false with p false.

  The values chosen for the minor clauses c_j may vary amongst the four cases.

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Restricted Inactive Clause Coverage

- Restricted Inactive Clause Coverage (RICC): For each $p \in P$ and each major clause $c_i \in C_p$, choose minor clauses $c_j, j \neq i$ so that c_i *does not* determine p.
  TR has four requirements for c_i:
  1. c_i evaluates to true with p true.
  2. c_i evaluates to false with p true.
  3. c_i evaluates to true with p false.
  4. c_i evaluates to false with p false.

  The values chosen for the minor clauses c_j must be the same in cases (1) and (2), and the values chosen for the minor clauses c_j must be the same in cases (3) and (4).

Infeasibility of logic coverage criteria

- Like many criteria, logic criteria can also be infeasible.
- RACC can be infeasible in many practical situations.
- There are logical expressions, which, interpreted practically, will make CACC feasible but, RACC infeasible.

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Feasibility: CACC vs. RACC, through an example

- System: A particular valve can be open or closed, system can be in two modes: "Operational" and "Standby".
- Two constraints on the system:
  - The valve must be open in "Operational" mode and closed in all the other modes.
  - The mode cannot be both "Operational" and "Standby" at the same time.
- Need to test for a certain action being taken only if the valve is closed and the system status is either "Operational" or "Standby".

Feasibility: CACC vs. RACC, through an example

- Predicate: p = valve is closed AND (system status is Operational OR system status is Standby).
- $p = a \wedge (b \vee c)$, where
  - a = The valve is closed,
  - b = The system status is Operational, and
  - c = The system status is Standby.
- Constraints can be written as the predicates $\neg a \equiv b$ and $\neg(b \wedge c)$.

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Feasibility: CACC vs. RACC, through an example

Constraints limit the feasible values in the truth table.

	a	b	c	$a \wedge (b \vee c)$	
1	T	T	T	T	violates constraints 1 and 2
2	T	T	F	T	violates constraint 1
3	T	F	T	T	
4	T	F	F	F	
5	F	T	T	F	violates constraint 2
6	F	T	F	F	
7	F	F	T	F	violates constraint 1
8	F	F	F	F	violates constraint 1

Only feasible rows are 3, 4 and 6.

Recap: CACC on $p = a \wedge (b \vee c)$

- In $p = a \wedge (b \vee c)$, for a to determine p, $b \vee c$ must be true.
- $b \vee c$ can be made true in three ways.
- CACC is satisfied for a being the major clause by choosing one TR from rows 1,2,3 and one from rows 5,6,7.

	a	b	c	$a \wedge (b \vee c)$
1	T	T	T	T
2	T	T	F	T
3	T	F	T	T
5	F	T	T	F
6	F	T	F	F
7	F	F	T	F

- CACC is feasible for the predicate p.

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Recap: RACC on $p = a \wedge (b \vee c)$

- Clause a is the major clause: Only three of the nine sets of test requirements that satisfy will satisfy RACC.
- Row 1 is paired with row 5, row 2 is paired with row 6 and row 3 is paired with row 7.

	a	b	c	$a \wedge (b \vee c)$
1	T	T	T	T
5	F	T	T	F
2	T	T	F	T
6	F	T	F	F
3	T	F	T	T
7	F	F	T	F

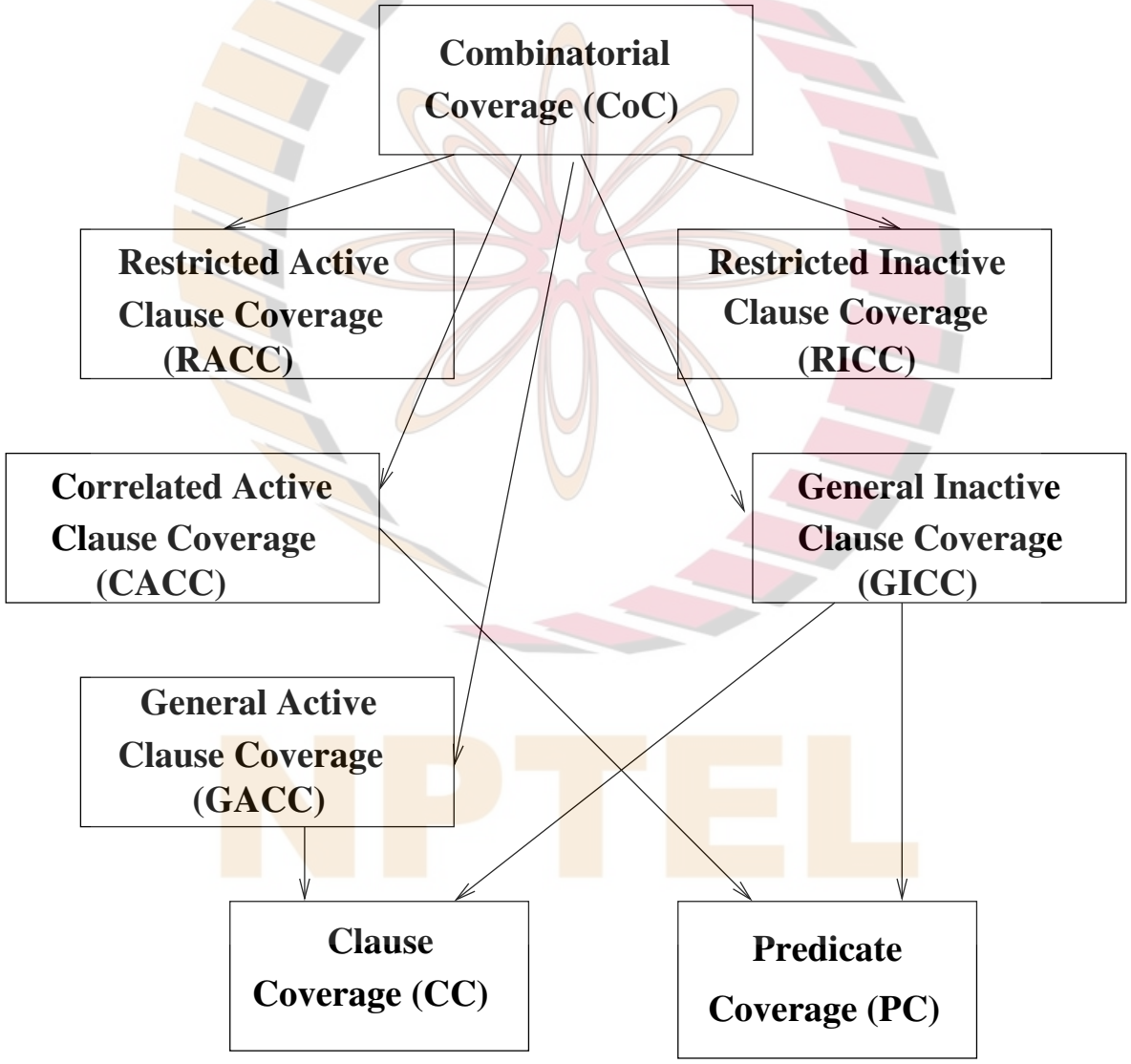
- RACC is infeasible for a in the predicate p.

Dealing with infeasibility

- We can ignore infeasible test requirements.
- We can consider counterparts of the requirements in a subsumed coverage criterion.
  - For e.g., if RACC with respect to a clause a is infeasible for a predicate p but CACC is feasible then, replace CACC with RACC.
- Similar to best effort touring for graphs.

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Subsumption relations among logic coverage criteria



Credits

Part of the material used in these slides are derived from the presentations of the book Introduction to Software Testing, by Paul Ammann and Jeff Offutt.

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